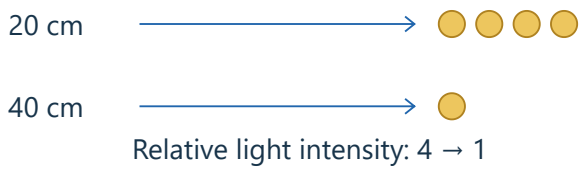
Photosynthesis practical

Set up fairly

1 Choose the variables



Relative light intensity: 4 → 1

Independent: lamp distance/light intensity.

Dependent: **oxygen production rate.**

Pondweed in sodium hydrogencarbonate solution; keep CO_2 supply, volume, plant and orientation constant.

2 Control temperature

Use a water bath/heat shield and check with a thermometer. The lamp can heat the water.

Keep background light steady. Use the same measurement period; avoid spills near electrics and hot-lamp burns.

Measure & calculate

3 Collect the data

Measure source-to-plant distance; allow the plant to **acclimatise.**

Collect oxygen volume over a set time using an inverted water-filled tube, or count bubbles.

Repeat at several distances; repeat each condition and calculate a mean. Investigate anomalies.

4 Calculate rate

Rate = gas volume/time. 3.6 cm^3 in 4 min = **0.90 cm^3/min.**

Bubble estimate: 48 in 2 min = **24 bubbles/min.**

Keep time/volume units consistent. More gas in a longer time is not necessarily a faster rate.

Evaluate evidence

5 Plot & explain

Plot rate on y-axis; intensity or **1/distance^2** on x-axis. Label units and use a suitable scale/best-fit curve.

Closer lamp → greater intensity; rate rises while light limits it. Another limiting factor can cause a plateau.

6 Limits & improvements

Unequal bubble sizes → count is approximate; measure **gas volume** instead.

Some oxygen dissolves or is used in respiration; check collector for gas leaks.

Repeats reduce random uncertainty; they do not fix heating, leaks or another uncontrolled variable.